

Project Title:

Blue-Green Deployment using AWS EC2 and Application Load Balancer

Project Overview:

This project demonstrates a **Blue-Green Deployment strategy** on AWS to achieve **zero-downtime application deployment**. Two identical environments (Blue and Green) were created using EC2 instances, and traffic was managed using an Application Load Balancer.

Objective

- Deploy application with **zero downtime**
- Enable **safe release of new versions**
- Provide **quick rollback capability**

Architecture Overview:

User

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Application Load Balancer

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Blue Environment Green Environment

(Version 1) (Version 2)

AWS Services Used:

- EC2 (Elastic Compute Cloud)
- Application Load Balancer (ALB)
- Target Groups
- AWS Services Used
- EC2 (Elastic Compute Cloud)
- Application Load Balancer (ALB)

Step-by-Step Implementation

Step 1: Launch Blue Environment

- Created an EC2 instance named **Blue-Server**
- Installed Apache web server
- Deployed Version 1 application:

Blue Environment - Version 1

Step 2: Launch Green Environment

- Created another EC2 instance named **Green-Server**
- Installed Apache
- Deployed Version 2 application:

Green Environment - Version 2

Step 3: Create Target Groups

- Created **Blue-TG** and registered Blue instance
 - Created **Green-TG** and registered Green instance
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Step 4: Create Application Load Balancer

- Configured ALB with HTTP listener (port 80)
 - Attached **Blue-TG** as default target group
 - Verified traffic routed to Blue environment
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Step 5: Perform Blue-Green Switch

- Modified ALB listener rule
 - Switched traffic from **Blue-TG → Green-TG**
 - Verified new version deployed without downtime
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Step 6: Rollback Strategy

- Traffic can be switched back to **Blue-TG** instantly in case of failure

Results:

Achieved zero downtime deployment

Successfully routed traffic between environments

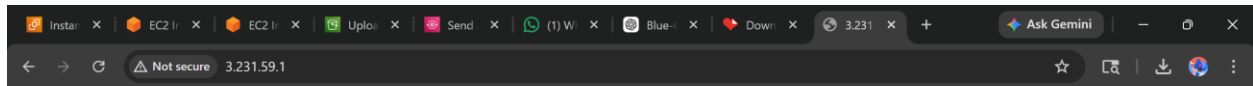
Enabled instant rollback capability

Improved deployment reliability

Key Learnings:

- Practical implementation of Blue-Green Deployment
- Working with **ALB and Target Groups**
- Managing EC2-based application environments
- Understanding production-level deployment strategies

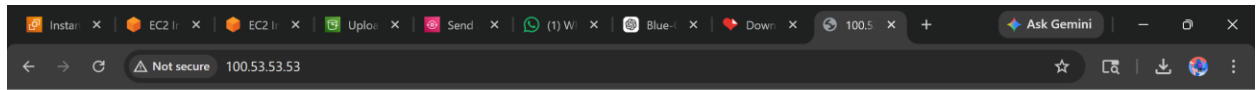
Final Outputs:



Green Environment - Version 2

New Version Deployed





Blue Environment - Version 1

Application Running

